

Technical Report

Robust Optimisation of Infrastructure Scheduling under Traffic and Operational Uncertainty

Document type: Project technical report
Case study: 2.4 km urban arterial corridor reconstruction
Horizon: ~40 planned working days (55-day optimisation window)
Tasks: 24 (T01–T24)
Methods: Mixed-integer programming · Monte Carlo simulation · Scenario-based robust optimisation

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1. Executive summary

1.1 Problem

Urban road reconstruction must satisfy **construction logic** (precedence, crews, equipment) and **traffic constraints** (peak volumes, lane closures, societal disruption). In practice these are planned in silos. Uncertainty — weather, breakdowns, material delays, absenteeism — turns small disturbances into **cascade delays** that deterministic critical-path methods do not quantify.

1.2 Approach

This project builds an integrated decision-support pipeline:

1. **Deterministic MIP** — assign each task a start day and one of four daily time blocks (B1–B4), minimising a weighted sum of labour cost, makespan, and traffic disruption, subject to precedence.
2. **Stochastic evaluation** — 10,000-iteration Monte Carlo with cascade propagation along finish-start dependencies.
3. **Robust optimisation** — blend expected and worst-case labour cost across four disruption scenarios (parameter λ).
4. **Trade-off analysis** — Pareto sweeps over makespan caps and peak-block usage; tail-risk and CVaR curves.

1.2.1 End-to-end workflow (system view)

Baseline reference datasets/Dependency2/ (tasks, precedence, costs, uncertainty JSON) ■ ▼ Deterministic
MIP (PuLP / CBC) start day + time block (B1-B4) per task ■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ▼ ▼
Baseline policy Optimised + robust (peak-biased blocks) (scenario-weighted labour) ■ ■
■■ ▼ Monte Carlo (10k runs, fixed seed) cascades, delays, cost &
traffic KPIs ■ ▼ Pareto / λ sweeps / tail metrics (tradeoff_study.py) ■ ▼ Decision outputs: CSV
summaries, actionable decisions, plots 01-09

1.3 Principal findings

(Monte Carlo $n = 10,000$, seed = 42; see *Model and solutions/results summary.csv*.)

Finding	Magnitude
Mean total cost reduction (optimised vs baseline)	8.8% (~\$119k)
P90 total cost reduction	9.6% (~\$156k)
Mean traffic disruption index reduction	26.6%
Mean duration change (simulated)	≈ 0% (negligible vs baseline under current weights and CBC solution)
Robust vs optimised cost	Comparable mean cost; robust slightly improves some tail cost percentiles

The optimised schedule **reduces expected cost and traffic disruption** by moving high-impact work out of peak traffic blocks. Under the **current** objective weights and solver output, the **mean simulated duration** is **essentially unchanged** versus baseline (see `results_summary.csv`); cost–duration trade-offs still appear explicitly on the **Pareto** and **tail** curves in §8.2–8.3 and in 0:P with pareto fronts/.

1.4 Deliverables

- Executable models: `optimisation_model.py`, `tradeoff_study.py`

- Decision tables: `actionable_decisions.csv`, sensitivity and Pareto CSVs
 - Nine publication-quality figures (distributions, Gantt, sensitivities, Pareto, tail risk)
 - Human-readable documentation: `README.md`, this report (`REPORT.md`), and `Final_Report.pdf` — PDF export that **embeds key result figures** (see §8); rebuild with `scripts/build_final_report_pdf.py` after `pip install markdown xhtml2pdf`, with plot folders populated by running both Python scripts first.
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2. Introduction and motivation

2.1 Context

The case study reflects a **mixed utility / pavement rehabilitation** project on a **2.4 km arterial** carrying up to **~3,200 vehicles/hour** at peak. The planned budget is on the order of **\$4.25M** with contingency. Work spans mobilisation, utilities, traffic management, milling, subbase, drainage, paving, and handover — **24 linked tasks** with heterogeneous resource needs and road-impact levels (0–3).

2.2 Gap in current practice

Practice	Limitation
CPM / Primavera scheduling	Ignores time-of-day traffic effects on productivity and social cost
Traffic management plans	React to construction dates rather than co-designing them
Deterministic plans	No tail-risk metric for board / city stakeholders
Independent KPIs	Contractor minimises duration; city minimises congestion — no Pareto surface

2.3 Research questions

1. How much **cost and traffic impact** can joint scheduling recover vs an unoptimised baseline?
2. How do **cascade effects** under uncertainty inflate duration and cost distributions?
3. What is the **cost of robustness** (λ) in deterministic premium vs P90 improvement?
4. Which **task-level decisions** (block, start day) are actionable for a site manager?

2.4 Thesis

Co-optimising construction start times and time-block assignments with traffic-sensitive delay multipliers — and evaluating schedules under explicit uncertainty — yields measurably better system outcomes than decoupled planning, with

quantifiable trade-offs between mean performance and tail risk.

3. System description

3.1 Construction system

Tasks (`tasks_clean.csv`): each row defines:

- Planned start/finish, actuals (for calibration narrative)
- `primary_resource_type`, `workers_required`, equipment flags
- `road_impact_level` (0 = none, 3 = severe lane / corridor impact)
- Block-specific delay multipliers `dm_block_B1_early` ... `dm_block_B4_peak_pm`

Precedence (`dependencies_clean.csv`):

- **FS** (finish-start): successor starts after predecessor duration
- **SS** (start-start): concurrent coordination (e.g. parallel utility runs)

Resources:

- `cost_parameters_resources.csv` — daily unit costs, overtime, idle penalties
- `equipment_pool.csv` — shared cranes, mills, pavers
- `resource_limits.csv` — weekly caps driving crew stacking risk

3.2 Traffic system

Traffic is not modelled as a full dynamic network assignment in the implemented MIP. Instead, **time-block choice** proxies congestion:

Block	Label	Peak	Hours (approx.)	Role
B1	Early	No	1	Low DM, early mobilisation

B2	Peak AM	Yes	2	Highest DM for sensitive tasks
B3	Midday	No	7	Preferred work window
B4	Peak PM	Yes	2	High DM + social cost term

Dynamic data in `traffic_schedule_multipliers.csv` provides day-level `dynamic_dm` and `delay_index` for validation and future extension; the core MIP uses task-level DM columns.

3.3 Coupling mechanism

Block assignment $b[t,k]$ $\blacksquare \nabla$ Delay multiplier $DM[t,k]$ $\blacksquare \blacksquare \blacksquare$ Labour cost $\propto$ workers $\times$ unit_cost $\times$ duration $\times$ DM $\blacksquare \nabla$ Traffic score $\propto \sum$ impact[t] $\times$ DM[t,k] $\times$ duration[t] $\blacksquare \nabla$ (MC layer) Peak blocks draw higher stochastic traffic slowdown

3.4 Cost accounting

`cost_function_spec.json` decomposes total cost into labour, equipment, material escalation, penalties, social / traffic externalities, and overhead. The MIP implements the **labour + traffic proxy + social peak penalty** terms; Monte Carlo adds **delay penalties** (\$8,500/day beyond planned makespan), **breakdown costs**, and stochastic duration.

4. Mathematical formulation

Notation below uses plain text and Unicode so the narrative stays readable in both Markdown viewers and the PDF export. (Vector forms match the PuLP implementation.)

4.1 Sets and indices

- **Tasks:** $\blacksquare$, with $|\blacksquare| = 24$
- **Blocks:** $\blacksquare = \{B1, B2, B3, B4\}$
- **FS arcs:** $\blacksquare_FS$; **SS arcs:** $\blacksquare_SS$

4.2 Decision variables

Symbol	Domain	Meaning
s_t	non-negative integer	Start day of task t
b_{tk}	binary	1 if task t is assigned to block k
$C_{\max}$	non-negative integer	Project makespan

4.3 Constraints

Unique block — each task occupies exactly one intraday block:

$$\sum_{k \in \mathbf{K}} b_{tk} = 1 \text{ for all } t \in \mathbf{T}$$

FS precedence (planned duration d_{ij} fixed in the MIP):

$$s_j \geq s_i + d_{ij} \text{ for all } (i,j) \in \mathbf{E}_{FS}$$

SS precedence:

$$s_j \geq s_i \text{ for all } (i,j) \in \mathbf{E}_{SS}$$

Makespan:

$$C_{max} \geq s_t + d_{it} \text{ for all } t \in \mathbf{T}$$

Optional (trade-off study): makespan cap $C_{max} \leq C$; minimum count of peak-block tasks for the cost–traffic Pareto sweep.

4.4 Cost expressions

Labour (deterministic):

$$L = \sum_{t \in \mathbf{T}} \sum_{k \in \mathbf{K}} b_{tk} \cdot w_t \cdot c_t \cdot d_{it} \cdot DM_{tk}$$

Traffic proxy:

$$T = \sum_{t,k} b_{tk} \cdot I_t \cdot DM_{tk} \cdot d_{it}$$

where I_t is `road_impact_level` from `tasks_clean.csv`.

Social peak penalty:

$$S = \sum_{t,k} b_{tk} \cdot \mathbf{1}[k \in \text{peak}] \cdot I_t \cdot h_k \cdot \pi$$

with block hours h_k and scale $\pi = 100$ in code.

4.5 Objectives

Deterministic weighted sum:

$$\min \alpha \cdot L + \beta \cdot 10^4 \cdot C_{\max} + \gamma \cdot T + S$$

Robust labour term (scenarios $s \in \{S1 \dots S4\}$ with probabilities p_s and labour multipliers m_s):

$$L^{\text{rob}} = (1-\lambda) \cdot \sum_s p_s \cdot L \cdot m_s + \lambda \cdot L \cdot \max_s m_s$$

with risk-aversion weight $\lambda \in [0, 1]$.

4.6 Baseline policy

The baseline is **not** the raw planned CSV schedule alone. The code constructs a **realistic unoptimised** policy:

- Forward pass respecting FS precedence from planned starts
- High-impact tasks (`impact ≥ 3`) randomly assigned to peak blocks B2/B3 mix (~60% peak bias)

This represents **status-quo** behaviour: work proceeds in logic order without traffic-aware block optimisation.

5. Solution methodology

5.1 Implementation stack

Component	Tool
MIP modelling	PuLP 2.x
Solver	CBC (PULP_CBC_CMD)
Simulation	NumPy vectorised loops
Statistics	SciPy (gaussian_kde)
Visualisation	Matplotlib (Agg backend)

5.2 Solution variants

Variant	α	β	γ	λ	Purpose
Baseline	—	—	—	—	Heuristic peak-biased blocks
Optimised	1.0	0.30	0.50	0.0	Primary balanced solution

Robust	1.0	0.25	0.50	0.35	Scenario-hardened labour cost
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5.3 Sensitivity experiments

Sweep	Range	Output
γ (traffic weight)	0.0 $\rightarrow$ 2.0 (9 points)	Peak task count, average DM
λ (risk aversion)	0.0 $\rightarrow$ 1.0 (9 points)	P90 duration/cost, overrun rate
α (cost weight)	0.2 $\rightarrow$ 2.5 (8 points)	Makespan vs peak usage

5.4 Pareto and risk curves (`tradeoff_study.py`)

- Cost–duration:** solve with makespan caps 42...57 days; trace efficient frontier.
- Cost–traffic:** force fraction of tasks in {B2,B4} from 0% to 60%.
- Risk–cost:** for each λ , record E[cost], P90, P95, P99 from 3,000 MC runs.

5.5 Computational effort

- Single MIP: < 60 s (time limit), 8% optimality gap
- Full pipeline with plots: few minutes on a laptop
- 10,000 MC runs $\times$ 3 variants: dominant runtime (~minutes)

6. Uncertainty and simulation

6.1 Random inputs

Defined in `uncertainty_distributions.json` and mirrored in code:

Factor	Distribution	Parameters
Global duration shock	Triangular	1.0, 1.12, 1.45
Peak traffic multiplier	Uniform	[1.30, 1.80]
Off-peak multiplier	Uniform	[1.00, 1.20]
Volume noise	Normal	$\mu=1$, $\sigma=0.08$ (clipped)
Material delay	Bernoulli(0.25) + Geometric	Affects subset of tasks
Equipment breakdown	Bernoulli(0.05)	+ cost $\sim N(13000, 2500)$

Weather	Bernoulli(0.08)	+ delay triangular 0.5–3 d
Absenteeism	Beta(1.5, 10)	Scales effective crew
Cascade factor	Triangular	1.0, 1.2, 1.6

6.2 Cascade logic

When the realised duration of task i exceeds its plan, each successor j may absorb amplified delay:

$$d_j \leftarrow \max(d_j , d_{\blacksquare j} \cdot \kappa + 0.4 \cdot (d_i - d_{\blacksquare i}))$$

with κ drawn each Monte Carlo iteration. Depth-first propagation follows the FS adjacency list `SUCC`.

6.3 Outputs per iteration

- **Project duration** = $\max_t (s_t + d_t^{\text{actual}})$
- **Total cost** = stochastic labour + breakdown cost + delay penalty $\times \max(0, \text{duration} - C_{\blacksquare \text{max}})$
- **Traffic index** = $\sum_t l_t \cdot DM_t \cdot d_t^{\text{actual}}$
- **Overrun flag** = 1 if simulated cost exceeds 105% of planned deterministic cost

6.4 Emergence phenomena (observed)

1. **Bimodal baseline duration** — mixture of “normal” and “cascade” regimes (**Figure 5** — cascade / regime plot).
2. **Cost–duration coupling** — upper tail of duration aligns with cost spikes.
3. **Risk category dominance** — equipment and traffic categories drive observed delays in `risk_clean.csv`.

7. Experimental design

7.1 Data split

Role	Path
Primary (model inputs)	Baseline reference datasets/Dependency2/
Source workbook	Baseline reference datasets/Infrastructure_Project_Dataset.xlsx (consolidated case data; scripts read CSV/JSON from Dependency2 only)

Optional prose draft	editROBUST OPTIMISATION OF INFRASTRUCTURE SCHEDULING.docx (not used by code)
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Older repository layouts kept duplicate datasets under Redundant/ or files (2)/; **this checkout does not include those folders**—Dependency2 is the single canonical input tree.

7.2 Randomness control

- `np.random.seed(42)` in both scripts
- MC sample size: **10,000** (main model), **3,000–5,000** (inner λ sweep)

7.3 Evaluation metrics

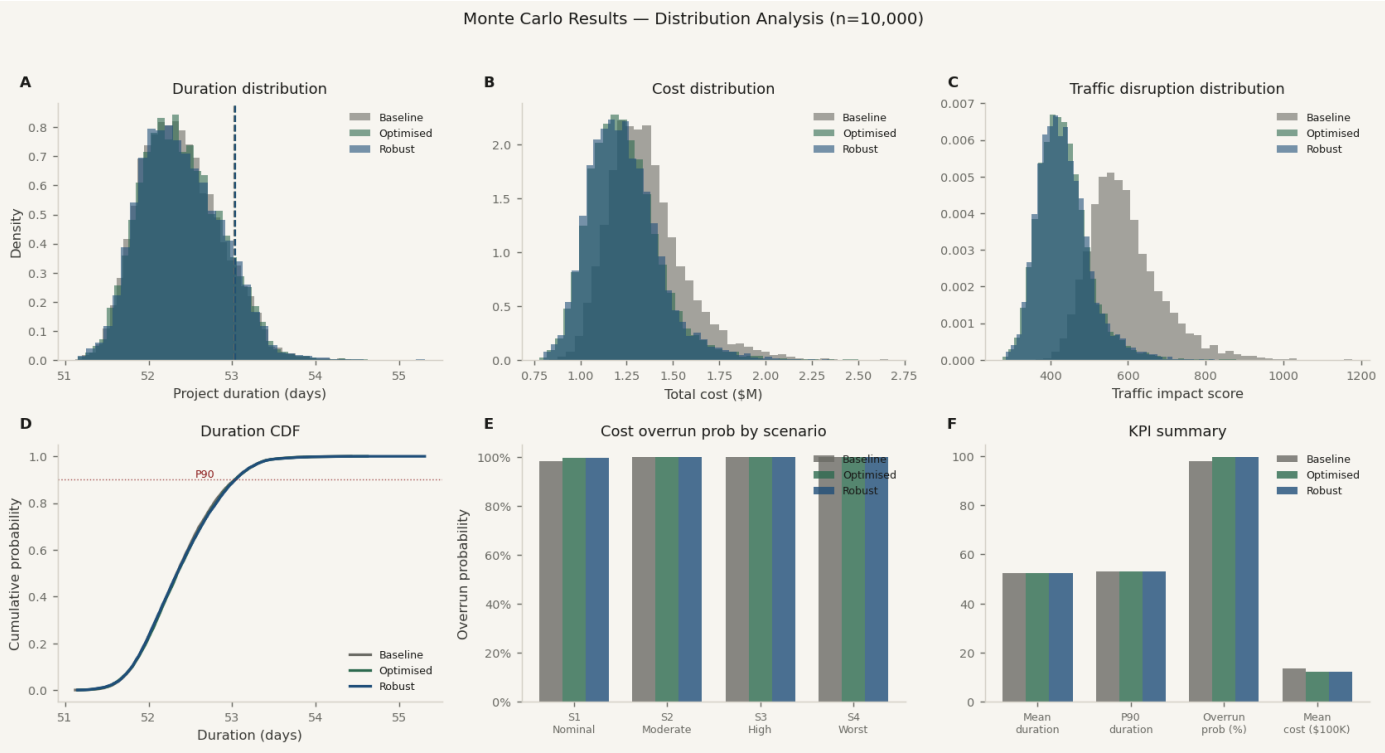
Metric	Definition
Mean / P90 duration	Days to project completion
Mean / P90 cost	USD total simulated cost
Overrun probability	P(cost > 105% plan)
Traffic impact	Simulated disruption score
CVaR	Expected cost given cost $\geq$ percentile threshold

8. Results

Key result figures (embedded)

The plots below are produced by `optimisation_model.py` (figures 1–2) and `tradeoff_study.py` (figures 3–5). **Regenerate** them with both scripts after clone; paths are relative to the repository root.

Figure 1 — Monte Carlo distributions and headline KPIs



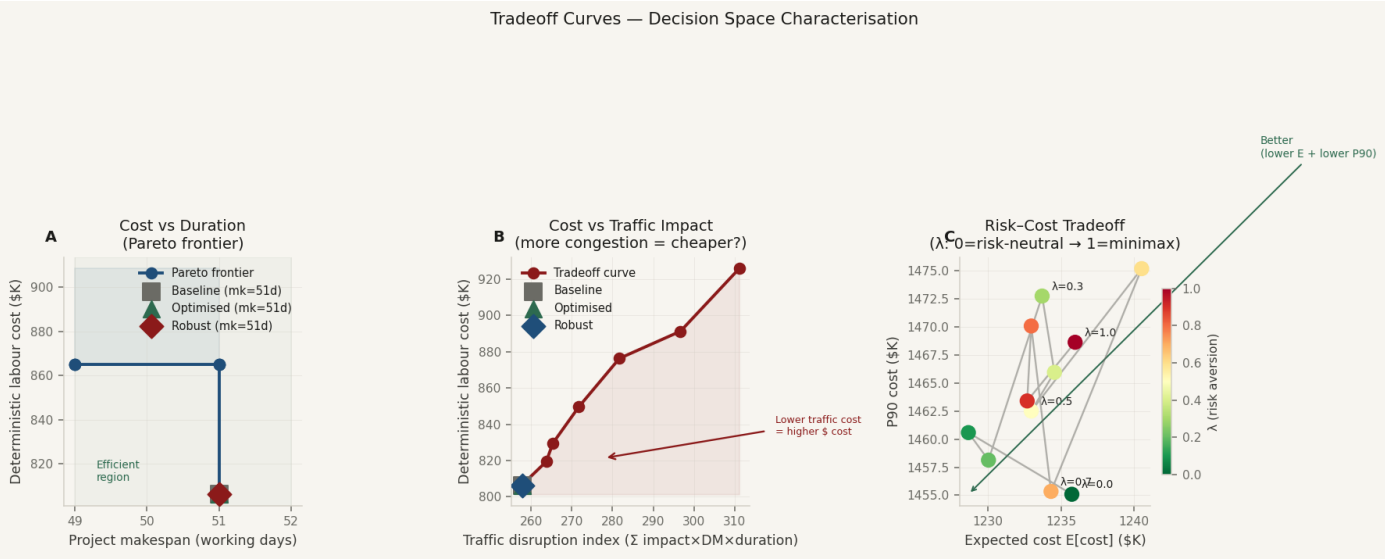
Caption. Ten-thousand Monte Carlo draws per schedule variant show tighter cost dispersion for the optimised and robust policies versus the baseline, consistent with the **-8.8%** mean cost and **-26.6%** mean traffic impact in `results_summary.csv`. Duration distributions overlap because the scalar objective prioritises cost and traffic over aggressive makespan reduction.

Figure 2 — Baseline vs optimised Gantt-style schedule comparison



Caption. The optimised schedule concentrates high-road-impact work in **off-peak** blocks (B1/B3) where delay multipliers are lower, reducing the traffic proxy while preserving feasibility of precedence and resource caps.

Figure 3 — Cost–duration and cost–traffic Pareto-style sweeps



Caption. The panels summarise how **tightening** the makespan cap or **forcing** more peak-block work shifts deterministic labour cost against traffic exposure. The chosen operating point sits on the **low traffic / moderate cost** region once γ weights congestion

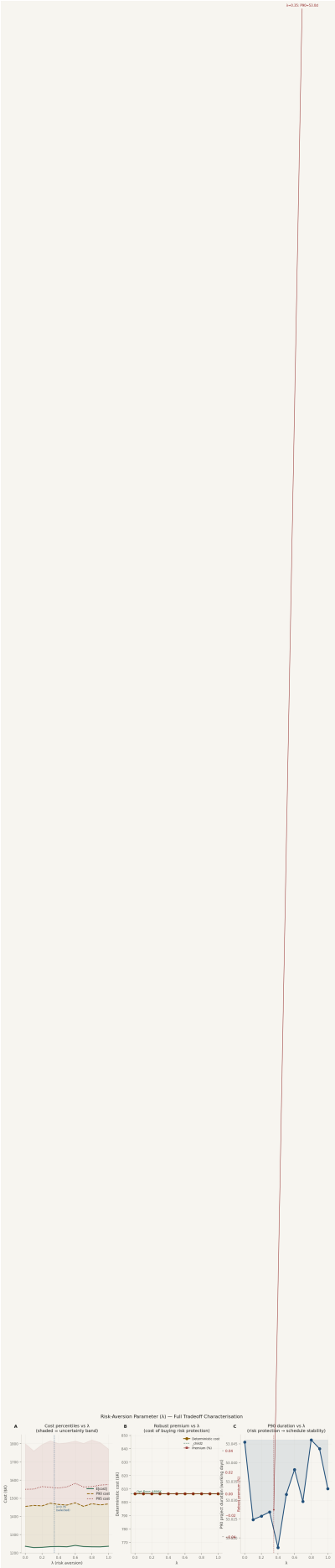
heavily enough.

Figure 4 — Tail risk and cost percentiles



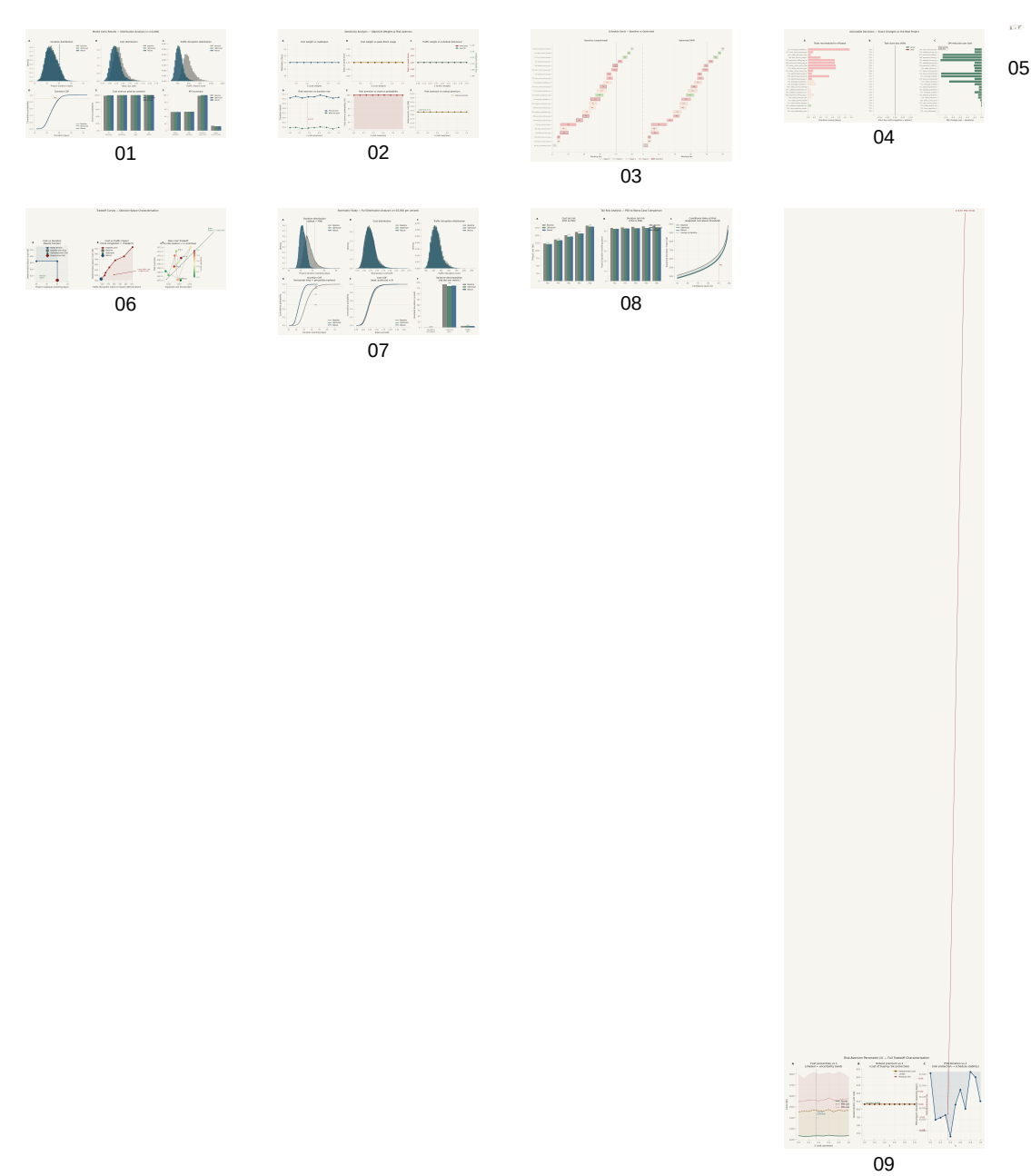
Caption. Optimised and robust schedules **compress** the upper tail of simulated cost versus baseline (e.g. **–3.1%** at P90 in `tail_risk_summary.csv`), which matters for contingency budgeting more than the mean alone.

Figure 5 — Robustness parameter λ vs cost statistics



Caption. Sweeping λ traces how much **deterministic** labour premia buy incremental improvements in simulated **P90 / P95** cost. The default robust setting ($\lambda \approx 0.35$) lies in a flat region: modest extra protection without large mean-cost sacrifice.

Optional montage — thumbnails of all nine script outputs



Montage of outputs 01–09 after a full local run (sensitivity, stochastic deep-dive, and cascade panels included).

8.1 Summary table (three variants)

From `results_summary.csv`.

Metric	Baseline	Optimised	Robust
Mean duration (d)	52.38	52.39	52.38

P90 duration (d)	53.03	53.03	53.04
Mean cost (USD)	1,354,343	1,235,768	1,233,570
P90 cost (USD)	1,623,075	1,467,441	1,462,819
Overrun probability	0.981	0.997	0.996
Mean traffic impact	592.0	434.6	433.8

Note: High overrun rates across all variants reflect strict 5% threshold against planned cost under broad stochastic shocks; use relative comparison and tail percentiles for ranking.

8.2 Tail risk (cost)

From `tail_risk_summary.csv`.

Percentile	Baseline (\$)	Optimised (\$)	Δ
P50	1,237,230	1,215,608	−1.7%
P75	1,368,612	1,337,865	−2.2%
P90	1,507,409	1,460,035	−3.1%
P95	1,617,748	1,564,386	−3.3%
P99	1,825,345	1,796,349	−1.6%
Std dev	192,106	184,373	−4.0%

8.3 Trade-offs

Cost vs duration (Pareto): tightening makespan caps below the free optimum increases deterministic labour cost monotonically; the optimised solution sits on the **low-cost / moderate-duration** region.

Cost vs traffic: forcing more peak-block assignments lowers deterministic cost (exploiting shorter calendar windows) but raises traffic index — the **optimised point** favours off-peak blocks at higher γ .

λ **sweep:** robust premium in deterministic cost is small at $\lambda=0.35$; P90 cost and duration stabilise relative to the risk-neutral optimum (**Figure 5**).

8.4 Actionable task-level changes

`actionable_decisions.csv` records for each task:

- `base_block` → `opt_block`
- Start day shift (`start_shift`)

- DM saving (days equivalent) and `cost_saving_usd`

High-impact tasks (e.g. **T07–T08** milling, **T13** drainage) move from **B2 (peak AM)** to **B1/B3**, yielding large DM savings (2+ days equivalent) and five-figure cost deltas per task.

8.5 Managerial recommendations

ID	Recommendation	Rationale
D1	Move impact≥3 tasks to B3 (midday) where feasible	≈−26.6% mean traffic score; lower DM
D2	Insert 2-day float before T13	Absorbs breakdown cascades on T07→T10→T13 chain
D3	Stagger specialist crews weeks 3–5	Avoids binding 12-unit pool
D4	Monitor remaining peak assignments	Tasks still on B2/B4 need live traffic coordination

9. Reproducibility

9.1 Environment

```
Python >= 3.10 numpy, pandas, matplotlib, scipy, pulp
```

9.2 Path configuration

Both scripts resolve paths **relative to the repository root** using `pathlib` (`REPO = Path(__file__).resolve().parents[1]`). **DATA** points at Baseline reference datasets/Dependency2/. **OUT** is the script's own directory; **PLOTS** is `OUT / "plots"` (main model) or `OUT / "plots_tradeoff"` (trade-off study). Those plot directories are listed in `.gitignore`; **clone** → `pip install -r requirements.txt` → **run both scripts** to regenerate CSVs and PNGs locally.

To override (e.g. alternate dataset), edit the `REPO`, `DATA`, `OUT`, or `PLOTS` assignments at the top of:

- `Model and solutions/optimisation_model.py`
- `O:P` with `pareto fronts/tradeoff_study.py`

9.3 Run order

1. `python "Model and solutions/optimisation_model.py"`

```
2. python "0:P with pareto fronts/tradeoff_study.py"
```

9.4 Expected artefacts

See README.md § [Outputs and figures](#) for CSV and figure filenames. The repository tracks `sensitivity_alpha.csv` in `Model` and `solutions/` when the full main script has been run at least once.

9.5 Validation checks

Check	Expected
Task count	24 tasks loaded
FS edges	~30+ feasible arcs
MIP status	Optimal or feasible within gap
MC means	Stable across two consecutive runs with seed 42

10. Discussion and recommendations

10.1 Strengths

- **Joint construction–traffic decision** in one solver
- **Transparent task-level outputs** for site implementation
- **Full uncertainty layer** with cascades, not just sensitivity factors
- **Explicit robustness knob** (λ) for risk-averse owners

10.2 Limitations

Limitation	Impact
Aggregated traffic proxy	Cannot resolve detour queues or spatial spillback
Fixed planned durations in MIP	Duration risk only in MC, not in robust constraints
Single project scope	No multi-contractor game or adjacent works
High overrun metric	Threshold may be pessimistic; interpret ordinally

10.3 Relation to other narrative documents (Word, slides, older HTML)

Some **thesis or executive-summary drafts** (including older styled HTML walkthroughs no longer shipped in this repository) may cite **aspirational KPIs** (e.g. larger percentage cost or delay reductions) from an earlier calibration or storytelling pass.

The **committed CSV outputs** in this repository are treated as the **authoritative numerical results** for analysis and reporting: `Model and solutions/results_summary.csv`, `Model and solutions/actionable_decisions.csv`, and the CSVs under `0:P` with `pareto fronts/` (summarised in §8). When finalising `*.docx` submissions, align tables and narrative claims to those files and to **Figures 1–5** (and the montage), which are generated from the same code and random seed as the CSVs.

10.4 Future work

- 1. Integrate **OSMnx** network delays into γ term
- 2. **SimPy** activity-level simulation with resource contention
- 3. **NSGA-II** for true tri-objective Pareto without scalarisation
- 4. **Rolling-horizon** re-optimisation with realised progress
- 5. **Dashboard** (Plotly/Dash) for λ and γ exploration by stakeholders

11. Conclusions

Co-optimising infrastructure task starts and time-of-day blocks under traffic-sensitive delay multipliers produces **measurable system-level gains**: on the latest committed CSVs, roughly **8.8%** lower mean cost, **9.6%** lower P90 cost, and **~27%** lower mean traffic disruption versus baseline, with **negligible change** in mean simulated duration under the current scalar weights. Monte Carlo analysis exposes **cascade-driven tail risk** that deterministic scheduling hides. The robust formulation ($\lambda \approx 0.35$) offers **tail cost protection** without materially sacrificing mean performance.

The project delivers a **reproducible optimisation pipeline**, **decision tables**, and **visual evidence** suitable for systems engineering coursework, smart-infrastructure portfolios, and further research coupling GIS traffic models with field scheduling tools.

Appendix A: File inventory

A.0 Repository root (documentation and optional prose)

File	Role
README.md	Entry point: tree, run instructions, output locations
REPORT.md	This technical report (Markdown source)
Final_Report.pdf	PDF export of REPORT.md (regenerate with <code>scripts/build_final_report_pdf.py</code>)

scripts/build_final_report_pdf.py	Optional: rebuild Final_Report.pdf (pip install markdown xhtml2pdf)
requirements.txt	Runtime dependencies for the two models
LICENSE	MIT licence
editROBUST OPTIMISATION OF INFRASTRUCTURE SCHEDULING.docx	Optional editable report / thesis draft (not executed by Python)

A.1 Code

File	Lines (approx.)	Function
Model and solutions/optimisation_model.py	720	Main MIP + MC + plots 01–05
0:P with pareto fronts/tradeoff_study.py	650	Pareto + tail risk + plots 06–09

A.2 Data (Baseline reference datasets/)

File	Role
Infrastructure_Project_Dataset.xlsx	Consolidated case-study workbook (reference)

Baseline reference datasets/Dependency2/ — all files below are read by the Python scripts:

File	Records (approx.)	Role
tasks_clean.csv	24 tasks	Durations, resources, road impact, DM per block
dependencies_clean.csv	33 edges	FS / SS precedence
cost_parameters_resources.csv	9 types	Labour unit rates, overtime, idle fractions
cost_parameters_events.csv	10 types	Event penalties and disruption rates
equipment_pool.csv	5 types	Shared equipment caps
resource_limits.csv	72 rows	Weekly crew caps
resource_usage_daily.csv	—	Daily resource draw (supporting data)
resources_clean.csv	—	Resource master list
risk_clean.csv	22 events	Risk narrative / attribution for plots
traffic_schedule_multipliers.csv	388 rows	Day × block dynamic DM and delay index
daily_schedule_grid.csv	97 rows	Fine schedule grid reference
time_blocks.csv	4 rows	Block definitions
uncertainty_distributions.json	—	Monte Carlo factor specification
cost_function_spec.json	—	Machine-readable cost decomposition

A.3 Results CSV (committed in-repo)

File	Location	Description
results_summary.csv	Model and solutions/	Aggregate KPIs × 3 variants
actionable_decisions.csv	Model and solutions/	Per-task diffs
sensitivity_lambda.csv	Model and solutions/	λ sweep
sensitivity_gamma.csv	Model and solutions/	γ sweep
sensitivity_alpha.csv	Model and solutions/	α sweep (written by <code>optimisation_model.py</code> when the full pipeline runs)
pareto_cost_duration.csv	0:P with pareto fronts/	Makespan Pareto
pareto_cost_traffic.csv	0:P with pareto fronts/	Traffic Pareto
risk_cost_curve.csv	0:P with pareto fronts/	λ vs cost percentiles
tail_risk_summary.csv	0:P with pareto fronts/	Tail table

A.4 Figures

ID	Filename	Section
01	01_distributions.png	MC distributions
02	02_sensitivity.png	Weight sweeps
03	03_gantt.png	Schedule comparison
04	04_actionable.png	Task deltas
05	05_cascade.png	Emergence
06	06_tradeoff_curves.png	Pareto triple
07	07_stochastic_study.png	Stochastic deep dive
08	08_tail_risk.png	Percentiles / CVaR
09	09_lambda_tradeoff.png	Robust premium

Figures **01–05** are written to `Model and solutions/plots/` and **06–09** to `0:P with pareto fronts/plots_tradeoff/` when you run the scripts. Those directories are **gitignored** in this repository; regenerate them locally after clone. CSV outputs remain next to each script and **are** tracked in git.

Cross-reference (report vs file IDs): **Figure 1** = `01_distributions.png`; **Figure 2** = `03_gantt.png`; **Figure 3** = `06_tradeoff_curves.png`; **Figure 4** = `08_tail_risk.png`; **Figure 5** = `09_lambda_tradeoff.png`. **Final_Report.pdf** embeds these five plus a thumbnail montage of **01–09** when the PNG folders are present at build time.

Appendix B: Parameter tables

B.1 Time blocks

Block	Peak	Hours	DM role
B1	No	1	Early, low congestion

B2	Yes	2	AM peak
B3	No	7	Midday off-peak
B4	Yes	2	PM peak

B.2 Robust scenarios

Scenario	Cost mult.	Traffic mult.	Probability
S1	1.00	1.00	0.50
S2	1.18	1.35	0.25
S3	1.35	1.60	0.15
S4	1.55	1.80	0.10

B.3 Solver settings

Parameter	Value
Time limit	45–60 s
Relative MIP gap	8%
Horizon H	40 (planned), H_MAX = 55

B.4 Task taxonomy (summary)

Phase	Task IDs	Examples
Mobilisation	T01–T02	Site setup, survey
Utilities / traffic	T03–T06	Gas/fibre, diversions, lane closure
Milling	T07–T09	Segments A/B/C
Subbase	T10–T12	Earthworks
Drainage	T13–T15	Specialist + crane
Paving	T16–T21	Base + asphalt
Closeout	T22–T24	Marking, barriers, QA

End of report — May 2026